

Brain-Apparatus Communication: A Journal of Bacomics

ISSN: 2770-6710 (Online) Journal homepage: www.tandfonline.com/journals/tbac20

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To cite this article: Jing Zhao, Qian Zhang, Xinrui Wang, Xueshuo Liu, Jiaxin Li, Fengjie Fan, Zhenhu Liang & Xiaoli Li (2024) An EEG dataset for studying asynchronous steady-state visual evoked potential (SSVEP) based brain computer interfaces, Brain-Apparatus Communication: A Journal of Bacomics, 3:1, 2418650, DOI: 10.1080/27706710.2024.2418650

To link to this article: https://doi.org/10.1080/27706710.2024.2418650

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Published online: 26 Oct 2024.

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BRAIN-APPARATUS COMMUNICATION: A JOURNAL OF BACOMICS
https://doi.org/10.1080/27706710.2024.2418650

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RESEARCH ARTICLE

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An EEG dataset for studying asynchronous steady-state visual evoked potential (SSVEP) based brain computer interfaces

Jing Zhao[a], Qian Zhang[a], Xinrui Wang[b], Xueshuo Liu[a], Jiaxin Li[a], Fengjie Fan[a], Zhenhu Liang[a] and Xiaoli Li[c,d]*

[a]Department of Electrical Engineering, Yanshan University, Qinhuangdao, China; [b]Suzhou Automobile Research Institute, Tsinghua University, Suzhou, China; [c]Guangdong Artificial Intelligence and Digital Economy Laboratory (Guangzhou), Guangzhou, China; [d]School of Automation Science and Engineering, South China University of Technology, Guangzhou, China

ABSTRACT

Compared with the commonly used synchronous brain-computer interface (BCI), the asynchronous BCI is a more flexible and natural way to control the real-world robotic devices. The major difficulty of building a robust asynchronous BCI lies in the discrimination between control states (CSs) and non-control states (NSs). This article presents an open-source 63-channel electroencephalogram (EEG) dataset of 24 subjects for asynchronous steady-state visual evoked potential (SSVEP)-BCI research. The data was recorded from an SSVEP based CS task and three different types of NS tasks, namely NS1, NS2 and NS3. The dataset was evaluated using three processes, including analysis of temporal waveform, amplitude spectrum and signal-to-noise (SNR), recognition of SSVEP frequencies, and classification between CS and NS. The dataset can be used to support studies of asynchronous classification, NS detection, and discrimination between different NSs.

ARTICLE HISTORY Received 2 January 2024; Accepted 6 October 2024

KEYWORDS Brain–computer interface (BCI); electroencephalogram (EEG); steady-state visual evoked potential (SSVEP); asynchronous; non-control state (NS)

CONTACT Xiaoli Li xiaolili@scut.edu.cn School of Automation Science and Engineering, South China University of Technology, Guangzhou 510641, China.

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1. Introduction

Brain-computer interface (BCI) provides a direct communication pathway between human brain and computer devices. In the current electroencephalogram (EEG)-based BCIs, steady-state visual evoked potential (SSVEP) is a paradigm widely used for control of robots because of its high information transfer rate (ITR) and low training demands [1, 2]. Several visual stimuli, corresponding to different commands to control a robot, are flickering at different frequencies on a user interface to evoke the SSVEP responses [2–5].

To classify the SSVEP signals, both synchronous methods and asynchronous methods are developed. The synchronous methods [6–10] are designed to identify only the frequency or phase components related to the flickering stimuli, while the asynchronous methods [11–15] also recognize the control state (CS) and non-control state (NS) in the SSVEP task. Compared with the synchronous method, the asynchronous method enables the participant to start and stop the control task (for SSVEP, fixating on the stimulus) at will, and is a more flexible and natural way to control the real-world robotic devices. But in an asynchronous BCI, as the participant is allowed to perform any mental activity other than fixating on stimuli during the non-control period, the major difficulty lies in the discrimination between the CSs and NSs [16].

Although there are several datasets [17–20] published for SSVEP-related research, NS trials in these datasets are usually recorded with small amount and short duration. Therefore, these datasets are more suitable for synchronous SSVEP research than for asynchronous research which requires a certain amount of NS data. The motivation of this article is to publish an open-source dataset with a large quantity and variety of NS data, and validate its effectiveness in offline classification study, to facilitate asynchronous SSVEP research. This study presents an open-source 63-channel EEG dataset of 24 subjects for asynchronous SSVEP-BCI research. The data was recorded from an SSVEP based CS task and three different types of NS tasks, namely NS1, NS2 and NS3. In the CS task, the subjects were asked to stare at a target stimulus for evoking SSVEP responses. In the three NS tasks, the subjects were asked to focus on the feedback image of a camera when the stimuli are not flickering, watch the white wall for a rest, and watch the feedback image when the stimuli are flickering continuously, respectively. For each subject, all the data was recorded within three hours and a short break was allowed between two neighboring sessions. Event markers were accurately synchronized to the recording of EEG data. This dataset contained the data epochs extracted from the recorded raw data without pre-processing. The data could be further pre-processed according to the researchers' needs. The dataset

supports in-depth study of artifact denoising, SSVEP recognition, NS detection, and discrimination between different NSs.

In this study, we validated the dataset using three evaluation processes, i.e., analysis of temporal waveform, amplitude spectrum and signal-to-noise (SNR), recognition of SSVEP frequencies, and classification between CS and NS. To evaluate the recognition performance of SSVEP frequencies, this study used three algorithms commonly used in SSVEP research, including filter-bank canonical correlation analysis (FBCCA) [8], task-related component analysis (TRCA) [7], and ensembled task-related component analysis (eTRCA) [7]. To evaluate the classification performance between the CS and NS, this study used two commonly used algorithms, i.e., individualized frequency band based optimized complex network (IFB-OCN) [12] and frequency specific framework using ensembled task-related component analysis (FS-eTRCA) [13]. An extension of the FS-eTRCA algorithm namely FS-eTRCA&FBCCA was also used for evaluation of discrimination performance between the CS and NS.

The remainder of this article is outlined as follows. The Methods Section introduces the subjects, stimulus presentation, experimental paradigm, and acquisition settings to record the dataset. The Data Records Section describes the related information of the dataset helpful for sharing. The Performance Evaluation Section presents the temporal, frequency and spatial features of the dataset and validates its effectiveness for asynchronous SSVEP classification. The Discussion Section discusses some issues that arose from this study and other notes, and gives suggestions for future work. The Conclusion Section summarizes the article. The dataset can be downloaded *via* the link provided by figshare [21].

2. Methods

2.1. Subjects

Twenty-four healthy individuals (9 females, aged 18–27 years, mean age: 22.96 years) took part in the experiment. All subjects had normal or corrected-to-normal vision and had no cognitive or psychological issues. Before the experiment, all subjects thoroughly understood the experimental procedures and signed written informed consent. The subjects were paid after the experiment. This study was reviewed and approved by the ethics committee of Qinhuangdao First Hospital.

2.2. Stimulus presentation

The experiment involved presenting visual stimuli on a 23.6-inch LCD monitor with a resolution of 1920 × 1080 pixels at 60 Hz. The stimulation

program was developed in MATLAB (MathWorks, Inc.) using the Psychophysics Toolbox Version 3 (PTB-3) [22]. Eight stimuli were modulated by sinusoidal modulation signals at frequencies of 8 Hz, 9 Hz, 10.5 Hz, 11 Hz, 12 Hz, 13 Hz, 14 Hz and 15 Hz, respectively. Figure 1 shows the stimulation interface involved the SSVEP task. The eight flickering stimuli were arranged in a 3×3 matrix, and a feedback image was displayed at the center of the interface.

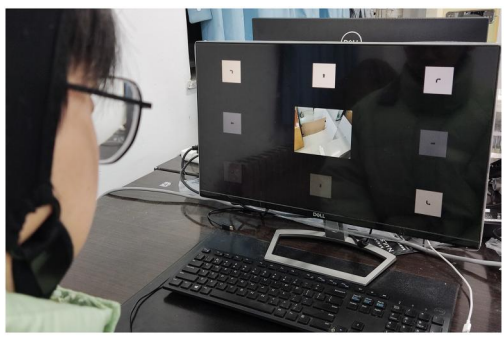
The control signal for each stimulus was generated by a sampled sinusoidal stimulation method [23]. At frequency f and phase θ, the stimulus sequence $s(f, \theta, i)$ can be generated by modulating the luminance of the screen.

$$s(f, \theta, i) = \frac{1}{2}\left\{1 + \sin\left[2\pi f\left(\frac{i}{RefreshRate}\right) + \theta\right]\right\} \tag{1}$$

where $\sin()$ generates a sine wave, i indicates the frame index in the sequence, and the refresh rate of the LCD monitor was 60 Hz. In the stimulus sequence, 0 denotes darkness and 1 denotes peak luminance. In this study, the phase θ was set to 0 for all the stimulation frequencies.

2.3. Experimental paradigm

Before the experiment, each subject was explained the experimental procedures and had a thorough understanding of the whole process. Meanwhile, the experimenter ensured reliability by monitoring the whole experimental process. The experiment was carried out in a laboratory environment without electromagnetic shielding. The subjects sat on an armchair 0.5 m away from the LCD monitor, and conducted an offline experiment to acquire the EEG data of the four states. In the experiment,



Target 1 8 Hz	Target 2 9 Hz	Target 3 10.5 Hz
Target 4 11 Hz	Camera image	Target 5 12 Hz
Target 6 13 Hz	Target 7 14 Hz	Target 8 15 Hz

(a) (b)

Figure 1. Stimulation interface. (a) Interface involved SSVEP stimulation. (b) Frequency values (phase value = 0) and location of stimuli.

the subject was instructed to perform the corresponding tasks, respectively. Figure 2 shows the experimental procedure of performing the four tasks. Between tasks, there was a break lasting several seconds or several minutes depending on subjects' preference.

1. CS task
   In the CS task, the subjects conducted a total of three sessions of the SSVEP experiment. Each session contained three blocks, each consisting of eight trials. Between sessions, the subjects were allowed to take a break lasting seconds or minutes as they wished. At the beginning of each trial, the user interface selected one stimulus as the target, following the order from lower frequency to higher frequency, and highlighted it with a static green square. After 1.5 s, the green square disappeared and the interface started to flicker the 8 stimuli for 4 s. During this 4-s period, the subjects were asked to stare at the target stimulus while avoiding possible body movements and eyeblinks. After that, the interface stopped the flickering of the stimuli, and the subjects took a rest for 0.5 s. The 4.5-s data epochs after the start of

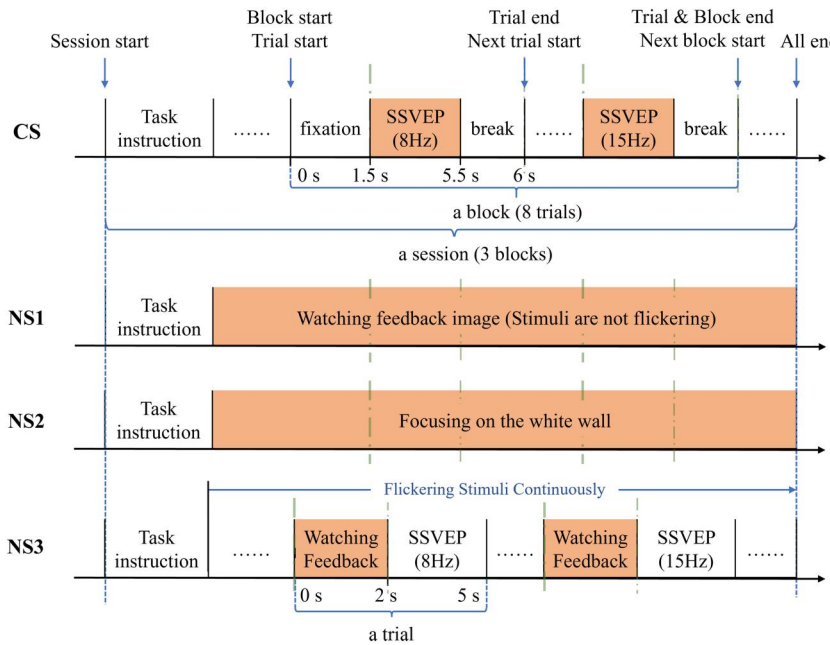


Figure 2. Experimental procedure of recording EEG data under different states. The orange color indicates the state we aim to record. The pairs of green-colored dashed lines indicate the data we extracted: the bolder line indicated the start of an extraction, and the finer line indicated the end of an extraction. For the NS1 and NS2 tasks, the data epochs were extracted using the same time as SSVEP focusing period in the CS task.

flickering were extracted for this dataset, consisting of a 4-s CS task and a 0.5-s break.

2. NS1 task
   The NS1 task was conducted using a similar stimulation program to the CS task, but the eight stimuli were not flickering during the whole task. The subjects were asked to watch the image feedback in the middle of the interface when the task started, taking no break during the NS1 task. Each subject conducted three blocks of NS1 tasks, each consisting of eight trials. The 4-s data epoch from 1.5 s to 5.5 s (0 indicates the start of a trial) of each trial were extracted as the NS1 task for this dataset.
3. NS2 task
   The NS2 task was conducted using the same stimulation program as the CS task. Different from the CS task, the NS2 task asked the subjects to watch the white wall and take a rest during the NS2 task. The 4-s data epochs from 1.5 s to 5.5 s (0 indicates the start of a trial) of each trial were extracted as the NS2 task for this dataset.
4. NS3 task
   The NS3 task was carried out using an asynchronous SSVEP paradigm. As shown in Figure 2, all the stimuli were flickering continuously for the whole task. The NS3 task required the subjects to watch the image feedback of surroundings for decisions making of subsequent commands. To acquire EEG data of the NS3 task, the subjects had to focus on the feedback image in the middle of the interface for 2 s. The interface then displayed a green square for 3 s to instruct the subjects to stare at the corresponding stimulus target. After that, the green indicator disappeared to start the next trial, and the subjects had to switch their gaze to the feedback image. Each subject conducted two sessions of the NS3 tasks, consisting of a total of 48 trials. The 2-s data epochs of focusing on the feedback image were extracted as the NS3 task for this dataset.

2.4. Data acquisition and preprocessing

This study used a 63 channel actiCHamp EEG system (Brain products, Germany) to record EEG data at a sampling rate of 5000 Hz. Electrodes were placed according to the international 10–20 system. The reference electrode was placed at Fz and the ground electrode was placed at FPz. Electrode impedances were kept below 50 kΩ during recording. The electrode placement of the EEG cap was shown in Figure 3.

Data epochs of the four tasks were extracted from the recorded raw data. For the CS task, each epoch contained a 4.5-s EEG data corresponding to

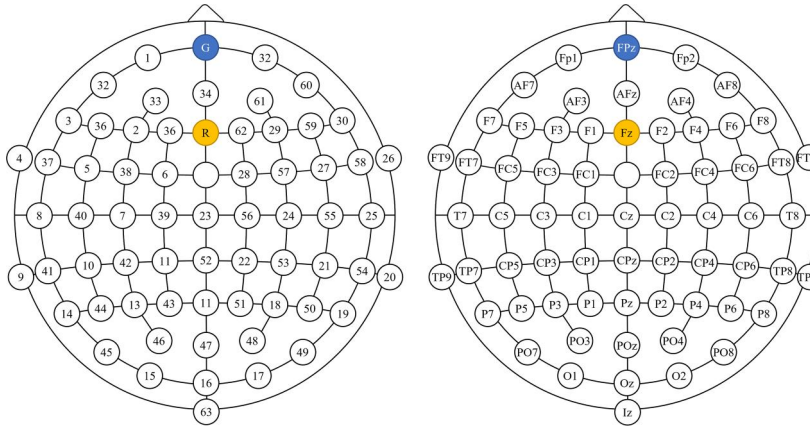


Figure 3. Electrode placement of the dataset.

Table 1. Detailed information of the dataset.

Parameters	Values
Number of subjects	24
Sampling rate	5000 Hz
Number of channels	63
Number of classes	4 (CS, NS1, NS2, NS3)

the 4-s SSVEP focusing and 0.5-s break; For the NS1 task, each epoch contained a 4-s EEG data recorded during watching the feedback image when the stimuli were not flickering; For the NS2 task, each epoch contained a 4-s EEG data recorded during gazing at the white wall. For the NS3 task, each epoch contained a 2-s EEG data recorded during gazing at the feedback image in the asynchronous SSVEP paradigm. Table 1 lists the detailed information of the dataset.

3. Data records

3.1. EEG data

Data were stored as double precision floating-point values in MATLAB. There are 24 folders named "Sxx" for each subject (i.e., S01, S02, …, S24). For each "Sxx" folder, the EEG data for the corresponding subject of the four tasks was saved as a ".mat" file respectively. The file naming rules were as follows:

sub_xx_data_yyy.mat

where "xx" was the subject number (01, 02, …, 24), "yyy" represented the control and non-control tasks including CS, NS1, NS2, and NS3. The

following data matrices were loaded in MATLAB by loading the corresponding ".mat" files.

- data_CS: consisting of 96 trials of EEG data recorded in the CS task. The data size was 8*63*22500*12 (target numbers, channel numbers, data samples and trial numbers). The eight targets correspond to the stimulation frequencies of 8 Hz, 9 Hz, 10.5 Hz, 11 Hz, 12 Hz, 13 Hz, 14 Hz and 15 Hz, respectively. The 22500 samples consisted of a 4-s SSVEP focusing and a 0.5-s break.
- data_NS1: consisting of 24 trials of EEG data recorded in the NS1 paradigm. The data size was 63*20000*24 (channel numbers, data samples and trial numbers).
- data_NS2: consisting of 24 trials of EEG data recorded in the NS2 paradigm. The data size was 63*20000*24 (channel numbers, data samples and trial numbers).
- data_NS3: consisting of 48 trials of EEG data recorded in the NS3 task. The data size was 63*10000*48 (channel numbers, data samples and trial numbers).

Table 2 lists the detailed information of the data we provided for a single subject.

3.2. Subject information

Information for all subjects was listed in "Subjects' Info.xlsx" file. For each subject, there are three factors including "Subject Index", "Gender" and "Age."

3.3. Electrode location

The electrode location was listed in "Channel Loc.xlsx." The file contains two columns named "Channel Index" and "Channel Name." The "Channel Index" ranges from 1 to 63, corresponding to the channel order of EEG data. Each specific channel can be indexed according to these two factors.

4. Performance evaluation

To validate the proposed dataset, this study performed three evaluation processes, including the analysis of temporal waveform, amplitude spectrum

Table 2. Detailed information about the data of a single subject.

Type	# of trials	Date length of each trial (s)	Data size
CS	96	4.5	$8 \times 63 \times 22500 \times 12$
NS1	24	4	$63 \times 20000 \times 24$
NS2	24	4	$63 \times 20000 \times 24$
NS3	48	2	$63 \times 10000 \times 48$

and SNR, the recognition of SSVEP frequencies, and the classification between CS and NS. For performance evaluation, data epochs were extracted with data lengths from 1 s to 4 s with an interval of 1 s, and were then pre-processed by removing baseline, down-sampling to 250 Hz, and notch filtered at 50 Hz. This study adopted classification accuracy and ITR for evaluation of the dataset. ITR (bits/min) was estimated as follows:

$$ITR = \left(\log_2 N + P\log_2 P + (1-P)\log_2\left[\frac{1-P}{N-1}\right]\right)\frac{60}{T} \tag{2}$$

where N is the number of classes, P is the accuracy of target identification, and T is the time length consisting of cue-guided time, target viewing time and visual delay.

4.1. Analysis of temporal waveform, amplitude spectrum, and SNR

To evaluate the signal quality of the CS task, this study analyzed temporal waveform, amplitude spectrum and SNR of SSVEP across all subjects. Figure 4(a) shows the temporal waveform of averaged 12 Hz SSVEP from the Oz electrode across all subjects. For each subject, all the 12 trials of SSVEP signals of 12 Hz were averaged first. For better observation of the temporal characters, the averaged data were band-pass filtered between 7 Hz and 70 Hz. The filtered data was then averaged across all the subjects. As shown in Figure 4(a), the waveform of SSVEP response was a near-sinusoidal signal at 12 Hz, time-locked to the stimulus onset with a delay around 130–140 ms. Figure 4(b) shows the amplitude spectrum of the SSVEP response to the stimulation at 12 Hz. The spectrum was estimated by averaging all the SSVEP trials of 12 Hz across all the subjects. The amplitude peaks can be observed at 12 Hz and its harmonic frequencies (i.e., 24 Hz, 36 Hz, and 48 Hz). The amplitude of the fundamental and harmonic components was decreased with the increase of frequencies (fundamental: 1.9006, second harmonic: 0.2071, third harmonic: 0.1413, fourth harmonic: 0.0656). Figure 4(c) shows the scalp topographies of the SSVEP response at 12 Hz and its harmonic frequencies. The occipital area delivered the highest amplitude for all the four components. Figure 4(d) shows the average SNR of SSVEP at 12 Hz. In this study, SNR is defined according to reference [17] by computing the ratio of the amplitude at a given frequency f to the mean amplitude of signals within the 2 Hz neighboring frequency band $[f-1, f+1]$.

$$SNR = \frac{\left(\frac{2}{\Delta f}+1\right)Amp_f}{\sum_{x=f-1}^{f+1} Amp_x} \tag{3}$$

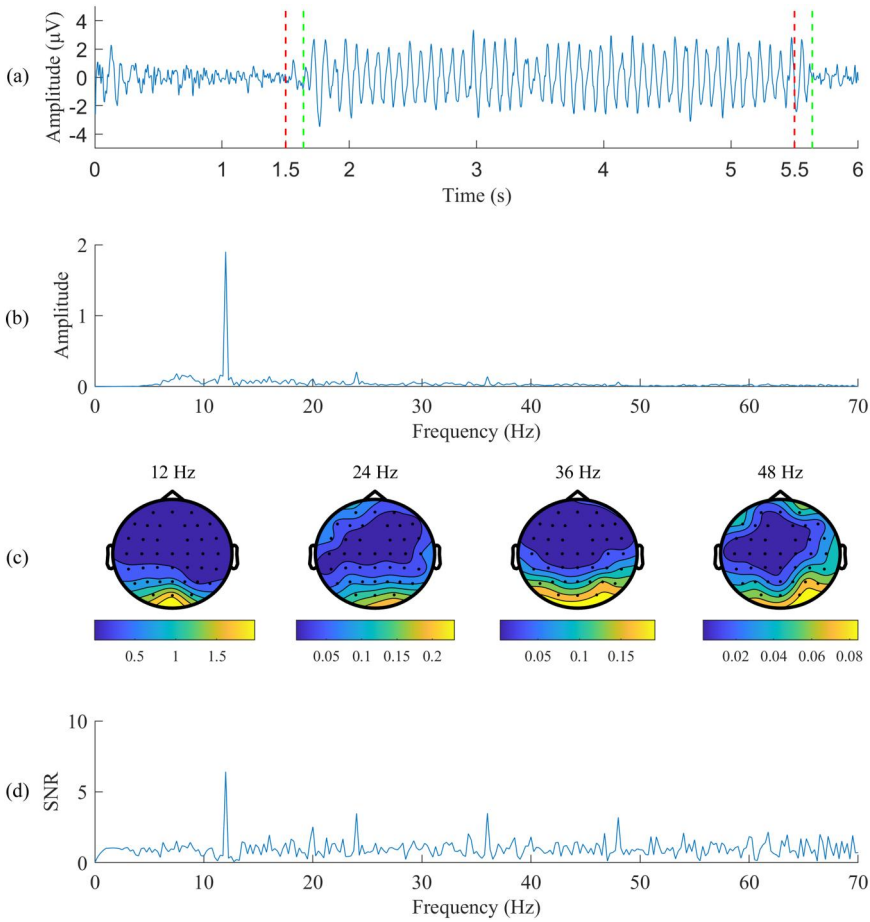


Figure 4. Signal quality analysis of the CS trials corresponding to the 12 Hz target. (a) Average temporal waveform from the Oz electrode. The red dashed lines indicate the onset and offset of the flickering. The green dashed lines indicate a visual delay of 140 ms. (b) Average amplitude spectrum. (c) Scalp maps at the fundamental, second, third, and fourth harmonic frequencies (12 Hz, 24 Hz, 36 Hz, and 48 Hz respectively). (d) Average SNR of 12 Hz SSVEP.

where Amp_f indicates the amplitude of frequency sepectra at f Hz, Δf indicates the interval between two neighbouring frequencies in the frequency spectra. In this study, Δf equals to 0.25. As shown in Figure 4(d), the SSVEP responses at the fundamental and harmonic frequencies delivered relatively high SNRs (fundamental: 6.4196, second harmonic: 3.4815, third harmonic: 3.4932, fourth harmonic: 3.2040), suggesting that the harmonic components provide useful information for detecting SSVEPs.

4.2. Recognition of SSVEP frequencies

To evaluate the recognition performance of SSVEP frequencies for the dataset, this study adopted three frequency recognition algorithms, i.e., FBCCA, TRCA and eTRCA, to classify the EEG data from the 8-class CS task for all the subjects. Nine electrodes including Pz, PO7, PO3, POz, PO4, PO8, O1, Oz and O2 were used for the following analysis. The three algorithms were evaluated using a 12-fold cross-validation, each fold consisting of 11 training trials and 1 test trial for each stimulation frequency. In this study, the number of sub-bands was set to 4, and the passbands of filter banks was set from m*8–2 Hz to 80 Hz. Here, $m = 1, 2, 3, 4$ indicates the index of filter bank. These algorithms recognized the target frequency according to the weighted sum of the correlation coefficients between the sub-band components and a template data. The FBCCA algorithm constructed the template using sinusoidal and cosine reference, while the TRCA and eTRCA algorithms constructed the templates based on the training set. Other parameters for these three algorithms were the same with references [7, 8].

Figure 5 shows the classification accuracies of the three algorithms with data lengths from 1 s to 4 s. The results showed that the average accuracies were improved with the increase of the data length. When the data length was set to 4 s, the FBCCA, eTRCA and TRCA algorithms obtained average accuracies of 93.49%, 79.47% and 73.00%, respectively. Among the three algorithms, the FBCCA algorithm achieved the highest accuracies regardless of data length, while the TRCA and eTRCA algorithms delivered larger inter-subject variability.

Figure 6 shows the ITRs of the three algorithms with data lengths from 1 s to 4 s. The results showed that all these algorithms obtained the highest ITRs with data length of 1 s. When the data length was set to 1 s, the FBCCA, eTRCA and TRCA algorithms obtained average ITRs of 43.8325 bits/min, 33.3243 bits/min and 29.1990 bits/min. As shown in Figure 6 (b)-(d), the eTRCA and TRCA algorithms delivered larger inter-subject variability than the FBCCA algorithm.

4.3. Classification between CS and NS states

In this study, we evaluated the classification performance of the dataset between the CS task and the three NS tasks. The data length was set to 1 s for the following analysis. For the CS task, 1-s data epochs were extracted from 0.14 s to 1.14 s for each trail (0 indicates the stimuli onset). For the NS tasks, the data trials were split into non-overlapping 1-s data epochs. Therefore, a total of 96 trials were acquired for each of the CS, NS1, NS2 and NS3 tasks. This study used the IFB-OCN, FS-eTRCA, and

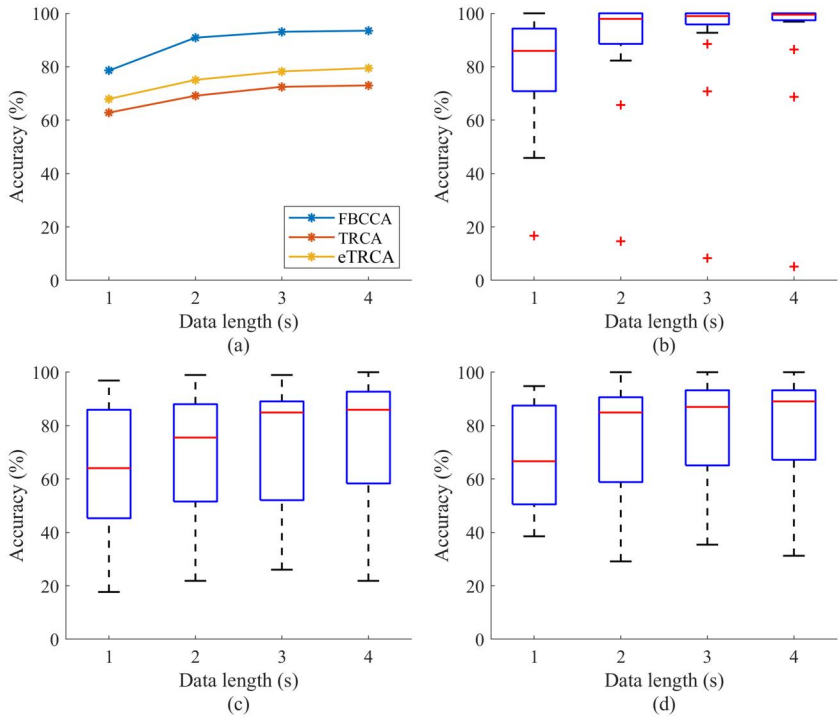


Figure 5. Classification accuracies of the FBCCA, TRCA and eTRCA algorithms with data lengths ranging from 1 s to 4 s in recognition of SSVEP frequencies. (a) Comparison of the accuracy between different algorithms. (b) Boxplot of the classification result for the FBCCA algorithm. (c) Boxplot of the classification result for the TRCA algorithm. (d) Boxplot of the classification result for the eTRCA algorithm.

FS-eTRCA&FBCCA algorithms to evaluate the classification performance between these tasks.

The IFB-OCN algorithm discriminated the CS and NS using the EEG data from a single channel at frontal area [12]. It divided the pre-processed data into multiple sub-bands, and selected the first three bands with the highest accuracies for feature extraction. The OCN features were extracted for each sub-band and were integrated for the final decisions. This study selected channel Fp1 and used a linear support vector machine (SVM) classifier [24] for classification. The FS-eTRCA algorithm extracted the correlation coefficient features between EEG data and the templates for the eight stimuli using the eTRCA algorithm. It identified the stimulation frequency by selecting the maximum coefficient, and trained linear discriminant analysis (LDA) [25] classifiers to discriminate between the corresponding CS and the NS. The FS-eTRCA&FBCCA algorithm was developed based on FS-eTRCA and adopted an additional FBCCA

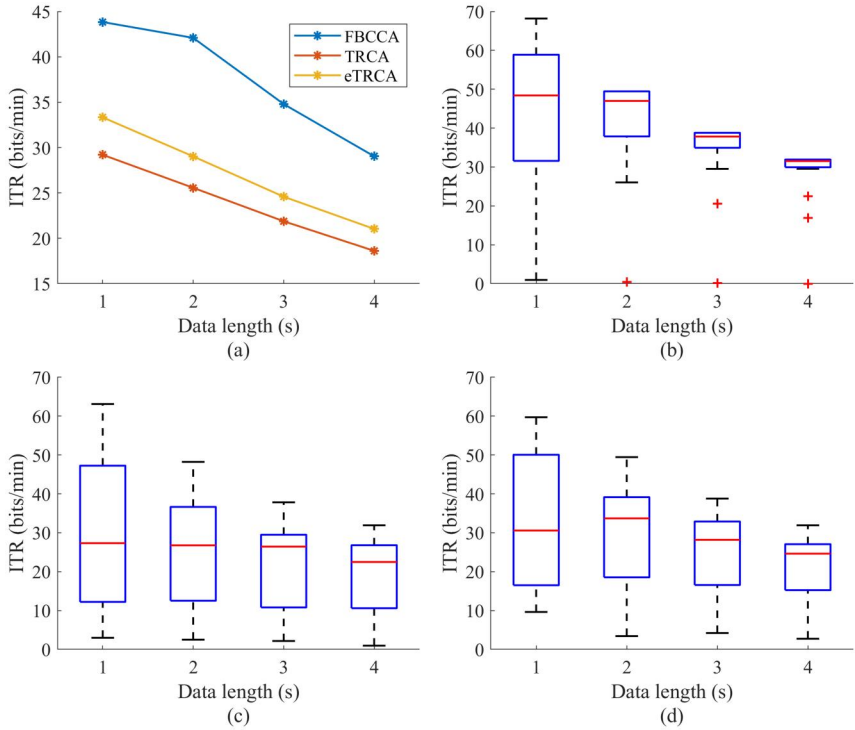


Figure 6. ITRs Of the FBCCA, TRCA and eTRCA algorithms with data lengths ranging from 1 s to 4 s in recognition of SSVEP frequencies. (a) Comparison of the ITR between different algorithms. (b) Boxplot of the ITR for the FBCCA algorithm. (c) Boxplot of the ITR for the TRCA algorithm. (d) Boxplot of the ITR for the eTRCA algorithm.

algorithm for feature extraction. The correlation coefficients extracted by eTRCA and FBCCA were combined as the final features for classification.

The classification performance of the three algorithms were evaluated using a 12-fold cross-validation. Different from the evaluation process of recognition of SSVEP frequencies, this evaluation trained and tested the algorithms using EEG data from the CS task and each of the three NS tasks, respectively. The IFB-OCN algorithm classified two classes of the CS and NS, while the FS-eTRCA and FS-eTRCA&FBCCA algorithms discriminated nine classes, including a NS and eight stimulation frequencies.

Figure 7 shows the classification accuracies of the three algorithms for discriminating between the CS and the three NSs. The results showed that the IFB-OCN algorithm achieved average accuracies of 70.51%, 72.35% and 69.34% in separating the CS with NS1, NS2 and NS3, respectively. The FS-eTRCA algorithm obtained an average accuracy of 67.64%, 66.47% and 65.34% in classifying nine classes including eight stimulation frequencies and a NS. The FS-eTRCA&FBCCA algorithm outperformed

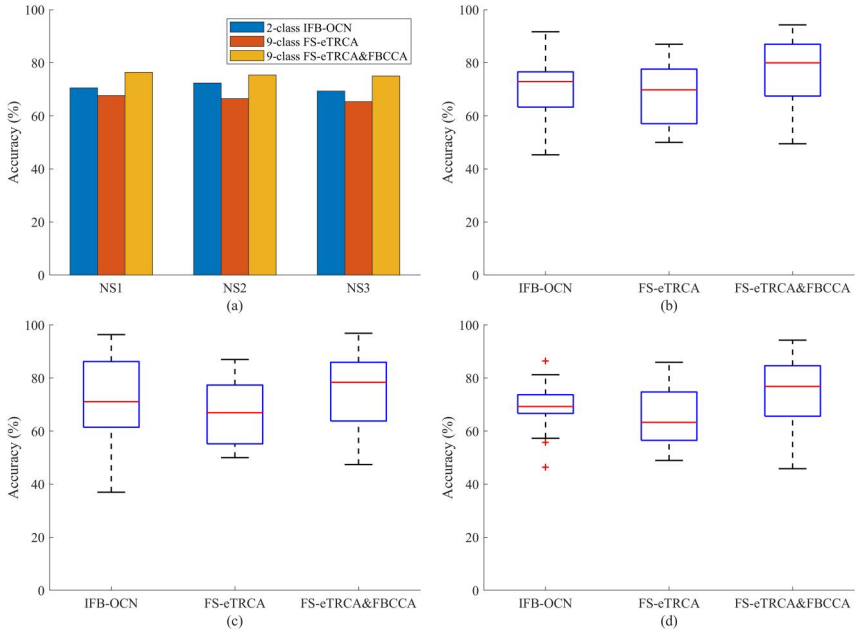


Figure 7. Classification accuracies of the IFB-OCN, FS-eTRCA and FS-eTRCA&FBCCA algorithms with data length of 1 s in classification between CS and NS. (a) Comparison of the classification accuracy between different algorithms. (b) Boxplot of the discrimination accuracy between the CS and NS1. (c) Boxplot of the discrimination accuracy between the CS and NS2. (d) Boxplot of the discrimination accuracy between the CS and NS3.

the FS-eTRCA by incorporating the FBCCA features and achieved average accuracies of 76.39%, 75.35% and 75.02%.

5. Discussions

As the existing SSVEP-related datasets usually record a small amount of NS trials with short duration, the motivation of this article is to publish an open-source dataset with large quantity and variety of NS data to facilitate the asynchronous SSVEP research. This dataset incorporates SSVEP data and three other types of NS data, providing rich potential for further exploration. Firstly, the dataset can be used for synchronous SSVEP research. The CS data is recorded corresponding to eight stimulation frequencies, and is suitable to investigate the algorithms for feature extraction, parameter optimization and classification [26]. Secondly, the dataset can be used for discrimination studies between CS and NS. In addition, the three types of NS data benefit the robustness validation of the discrimination algorithms. Thirdly, the dataset facilitates the investigation of relationships between different CS and NS.

However, there are still two limitations to our present work. On one hand, the influence factors in subject recruitment such as age group and education background are not yet considered in this dataset. Larger sample size and diverse of subjects will be recruited in our future data collection works. On the other hand, the experimental paradigm of this dataset is designed mainly for offline asynchronous studies. In these studies, the EEG data separately acquired from the CS and NS trials can be used for investigation of features and algorithms. But for online studies requiring continuous acquisition of all the states, this dataset is currently not suitable.

To verify the data quality of this dataset, SSVEP recognition was conducted using three methods with a time range from 1 s to 4 s. The FBCCA algorithm outperformed the TRCA and eTRCA algorithms in terms of accuracy (Figure 6) and ITR (Figure 7). One possible reason for this is that the stimulation phases were the same for all stimuli in this dataset. As the TRCA and eTRCA algorithms utilized both frequency and phase information for SSVEP recognition, a lack of phase variance reduced their classification performance. This also holds true for the classification between CS and NS, where the FS frame based on FBCCA and eTRCA outperformed the eTRCA-based FS frame.

In our future work, the dataset will be improved in the following aspects. Firstly, we will enlarge data recordings from a broader range of subjects to create a more generalized dataset. Additional factors will be considered during subject recruitment, such as age, education background and gender. Secondly, we will incorporate data recordings from different days of the same subject to facilitate investigation of the intra-individual variability. Thirdly, our future work will design experimental protocols for continuous acquisition of all CS and NSs, and publish dataset for evaluation of algorithms' online performance.

6. Conclusion

The presented study introduces a dataset for studying asynchronous SSVEP-based BCI. This dataset includes four distinct data types: CS data based on SSVEP at frequencies of 8, 9, 10.5, 11, 12, 13, 14 and 15 Hz, as well as three types of NS data. Through analysis of temporal waveforms, amplitude waveforms, SNR in the frequency domain, and spectral amplitude distribution in the spatial domain, we validated the SSVEP signal quality of our dataset. Moreover, the results of frequency recognition using FBCCA, TRCA, and eTRCA methods demonstrated the effectiveness of this dataset in SSVEP classification, and the results of state classification using IFB-OCN, FS-eTRCA and FS-eTRCA&FBCCA methods demonstrated effectiveness of this dataset in discrimination between CS and NS.

Disclosure statement

The authors report there are no competing interests to declare.

Funding

This research was supported by the National Key R&D Program of China (2022YFE0140400), National Natural Science Foundation of China (62376241), and S&T Program of Hebei (236Z2001G).

Data availability statement

The data that support the findings of this study are openly available in figshare at https://doi.org/10.6084/m9.figshare.24906300.v3, reference number [21].

References

1 Wang Y, Gao X, Hong B, et al. Brain-computer interfaces based on visual evoked potentials. IEEE Eng Med Biol Mag. 2008;27(5):64–71. doi: 10.1109/MEMB.2008.923958.
2 Zhao J, Li W, Li M. Comparative study of SSVEP- and P300-based models for the telepresence control of humanoid robots. PLoS One. 2015;10(11):e0142168. doi: 10.1371/journal.pone.0142168.
3 Chen X, Chen Z, Gao S, et al. Brain–computer interface based on intermodulation frequency. J Neural Eng. 2013;10(6):066009. doi: 10.1088/1741-2560/10/6/066009.
4 Ortner R, Guger C, Prueckl R, et al. SSVEP based brain-computer interface for robot control. In: Miesenberger K, Klaus J, Zagler W, Karshmer A, editors. Computers helping people with special needs. Berlin (Heidelberg): Springer; 2010. p. 85–90.
5 Güneysu A, Akin HL. An SSVEP based BCI to control a humanoid robot by using portable EEG device. Annu Int Conf IEEE Eng Med Biol Soc. 2013;2013:6905–6908. doi: 10.1109/EMBC.2013.6611145.
6 Yao H, Liu K, Deng X, et al. FB-EEGNet: a fusion neural network across multi-stimulus for SSVEP target detection. J Neurosci Method. 2022;379:109674. doi: 10.1016/j.jneumeth.2022.109674.
7 Nakanishi M, Wang Y, Chen X, et al. Enhancing detection of SSVEPs for a high-speed brain speller using task-related component analysis. IEEE Trans Biomed Eng. 2018;65(1):104–112. doi: 10.1109/TBME.2017.2694818.
8 Chen X, Wang Y, Gao S, et al. Filter bank canonical correlation analysis for implementing a high-speed SSVEP-based brain-computer interface. J Neural Eng. 2015;12(4):046008. doi: 10.1088/1741-2560/12/4/046008.
9 Wong CM, Wan F, Wang B, et al. Learning across multi-stimulus enhances target recognition methods in SSVEP-based BCIs. J Neural Eng. 2020;17(1):016026. doi: 10.1088/1741-2552/ab2373.
10 Pan Y, Chen J, Zhang Y, et al. An efficient CNN-LSTM network with spectral normalization and label smoothing technologies for SSVEP frequency recognition. J Neural Eng. 2022;19(5):056014. doi: 10.1088/1741-2552/ac8dc5.

11 Pan J, Li Y, Zhang R, et al. Discrimination between control and idle states in asynchronous SSVEP-based brain switches: a pseudo-key-based approach. IEEE Trans Neural Syst Rehabil Eng. 2013;21(3):435–443. doi: 10.1109/TNSRE.2013.2253801.

12 Zhang W, Zhou T, Zhao J, et al. Recognition of the idle state based on a novel IFB-OCN method for an asynchronous brain-computer interface. J Neurosci Methods. 2020;341:108776. doi: 10.1016/j.jneumeth.2020.108776.

13 Ke Y, Du J, Liu S, et al. Enhancing detection of control state for high-speed asynchronous SSVEP-BCIs using frequency-specific framework. IEEE Trans Neural Syst Rehabil Eng. 2023;PP:1405–1417. doi: 10.1109/TNSRE.2023.3246359.

14 Wang R, Zhou T, Li Z, et al. Using oscillatory and aperiodic neural activity features for identifying idle state in SSVEP-based BCIs reduces false triggers. J Neural Eng. 2023;20(6):066032. doi: 10.1088/1741-2552/ad1054.

15 Zhang D, Huang B, Wu W, et al. An idle-state detection algorithm for SSVEP-based brain-computer interfaces using a maximum evoked response spatial filter. Int J Neural Syst. 2015;25(7):1550030. doi: 10.1142/S0129065715500306.

16 Nagel S, Spüler M. Asynchronous non-invasive high-speed BCI speller with robust non-control state detection. Sci Rep. 2019;9(1):8269. doi: 10.1038/s41598-019-44645-x.

17 Wang Y, Chen X, Gao X, et al. A benchmark dataset for SSVEP-based brain–computer interfaces. IEEE Trans Neural Syst Rehabil Eng. 2017;25(10):1746–1752. doi: 10.1109/TNSRE.2016.2627556.

18 Liu B, Huang X, Wang Y, et al. BETA: a large benchmark database toward SSVEP-BCI application. Front Neurosci. 2020;14:627. doi: 10.3389/fnins.2020.00627.

19 Chevallier S. 2023. sylvchev/dataset-ssvep-exoskeleton [Internet]. [cited 2023 Dec 26]. Available from: https://github.com/sylvchev/dataset-ssvep-exoskeleton.

20 Vilic. Avi A, Dataset S. 2013. [dataset]. In: setzner [Internet]. [cited 2023 Dec 26]. Available from: https://www.setzner.com/avi-ssvep-dataset/.

21 Zhao J, Zhang Q, Wang X-R. YSU-an EEG dataset for studying asynchronous SSVEP-BCI [Internet]. figshare; 2023 [cited 2024 Oct 8]. Available from: https://figshare.com/articles/dataset/YSU-an_EEG_dataset_for_studying_asynchronous_SSVEP-BCI/24906300/3.

22 Brainard DH, Vision S. The psychophysics toolbox. Spat Vis. 1997;10(4):433–436. doi: 10.1163/156856897X00357.

23 Chen X, Chen Z, Gao S, et al. A high-ITR SSVEP-based BCI speller. Brain-Computer Interfaces. 2014;1(3-4):181–191. doi: 10.1080/2326263X.2014.944469.

24 Chang C-C, Lin C-J. LIBSVM: a library for support vector machines. ACM Trans. Intell. Syst. Technol. 2011;2(3):1–27:27. doi: 10.1145/1961189.1961199.

25 Tharwat A, Gaber T, Ibrahim A, et al. Linear discriminant analysis: A detailed tutorial. AIC. 2017;30(2):169–190. doi: 10.3233/AIC-170729.

26 Zhang H, Zhou Q-Q, Chen H, et al. The applied principles of EEG analysis methods in neuroscience and clinical neurology. Mil Med Res. 2023;10(1):67. doi: 10.1186/s40779-023-00502-7.